Points 12.25/16 (76.56 %)

I have really appreciated the Latex. It was so much better to correct!

Next time: \linespread{1.2} or 1.5 even!

Aufgabe 1

t.s. *The 7th cyclotomic field has a trivial classgroup.*

Proof. Let $p = 7$ and let $K = \mathbb{Q}[T]/(\Phi_p\mathbb{Q}[T])$ be the 7-th cyclotomic field. (I guess this is the 7th cyclotomic polynomial then.) Since $\zeta \in \mathbb{Z}_K$ is a primitive element of $K/\mathbb{Q}$we get $K = \mathbb{Q}(\zeta)$. Fruthermore by Lemma 3.26 it holds that $\mathbb{Z}_K = \mathbb{Z}[\zeta]$ ok. Since p is prime, it holds that Φ_p is a monic irreducible Polynomial in $\mathbb{Q}[T]$, ok (Φ_n is irreducible $\forall n \in \mathbb{N}$) where a previous Lemma implies that $\Phi_p = min_{K/\mathbb{Q}}(\zeta)$, thus $[K : \mathbb{Q}] = 7$ ($\phi(7)=6$) and $(1, \zeta, \ldots, \zeta^{p-1})$ form a $\mathbb{Q}$-Basis of the $\mathbb{Q}$-Vector space K. Furthermore $(1, \zeta, \ldots, \zeta^{p-1})$ forms a $\mathbb{Z}$-Basis of The $\mathbb{Z}$-Modul $\mathbb{Z}_K$. Then since $[\mathbb{Z}_K : \mathbb{Z}[\zeta]] = 1$, the assumptions of Satz 2.9 (maybe S.3.9) are satsified. Then $\triangle_p = + - p^{p-2} \equiv 0 \pmod{p}$.

Now let I be a sub$\mathbb{Z}_K$-Modul of K. ($K = Quot(\mathbb{Z}_K)$). Then there exists $y_1, \ldots, y_N \in K$, where $N < p$, such that

$$I = y_1\mathbb{Z}_K + \cdots y_N\mathbb{Z}_K.$$

Now we want to show that there exists a $a \in K$: such that $a\mathbb{Z}_K = I$.
(Then we can find a $b \in \mathbb{Z}$, such that $ab \in \mathbb{Z}_K$, which would imply that I is a principle ideal, hence $\mathbb{Z}_K$ would be A principle ideal domain and we could conclude.)
But, sorry Alessio. I have no idea how I can do that. (LoL) □ 0.25

DE : Modul
UK/US: module

Aufgabe 2

Let $m \in \mathbb{N}$ and let $\delta_1, \dots, \delta_m > 0$, such that $\delta_1 \cdots \delta_m = 1$. Further let $A = (a_{i,j})_{i,j} \in Mat_m(\mathbb{R})$, where $det(A) \in \{-1, 1\}$.

t.s. *There exists a* $(x_1, \dots x_m) \in \mathbb{Z}^m \setminus \{0\}$, *such that*

$$\left| \sum_{j=1}^{m} a_{ij} x_j \right| < \delta_i,$$

for each $i \in \{1, \dots, m\}$.

Proof. We define

$$S = \left\{ (x_1, \dots, x_m) \in \mathbb{R}^4 : \left| \sum_{i=1}^{m} a_{ij} x_j \right| \leq \delta_i \text{ for each } i \in \{1, \dots, m\} \right\} \subseteq \mathbb{R}^4. \quad \text{ok}$$

Then for any $x = (x_1, \dots, x_m) \in \mathbb{R}^4$, the following equivalences hold

$$\left| \sum_{i=1}^{m} a_{ij} x_j \right| \leq \delta_i \text{ for each } i \in \{1, \dots, m\} \Leftrightarrow |(A \cdot x)_i| \leq \delta_i \text{ for each } i \in \{1, \dots, m\}$$
$$\Leftrightarrow (A \cdot x)_i \in [-\delta_i, \delta_i] \text{ for each } i \in \{1, \dots, m\}$$

where $(A \cdot x)_i$ is the i-th coordinate of $A \cdot x \in \mathbb{R}^4$. ? Maybe you mean the i-th raw...
Now let $L : \mathbb{R}^4 \to \mathbb{R}^4$ be a linear map, given by $x \mapsto A \cdot x$. Then L is an automorphism on $\mathbb{R}^4$, ($A \in Gl_n(\mathbb{R}^4)$) and the above equivalence implies that

$$S = L^{-1}(U), \text{ where } U = \prod_{i=1}^{m} [-\delta_i, \delta_i] \quad \checkmark$$

Then it holds that $\text{vol}(U) \overset{\checkmark}{=} \prod_{i=1}^{m} (2\delta_i) \overset{\checkmark}{=} 2^m \prod_{i=1}^{m} (\delta_i) \overset{\checkmark}{=} 2^m$, since $\prod_{i=1}^{m} (\delta_i) = 1$ by assumption. Any linear automorphism on $\mathbb{R}^4$ is bounded and therefore continuous. Moreover, one can see that the partial derivatives of L^{-1} correspond to the column vectors of A^{-1}, which means that the partial derivatives are endomorphisms from $\mathbb{R}^4$ to $\mathbb{R}$ and are therefore continuous. So L^{-1} is totally differentiable and in particular

$$D_x L^{-1} = L^{-1} \text{ for every } x \in \mathbb{R}^4,$$

so change of variable provides us that

$$\text{vol}(S) \overset{\checkmark}{=} |\det(L^{-1})| \text{vol}(U) \overset{\checkmark}{=} \underbrace{\left| \frac{1}{\det(A)} \right|}_{=1} 2^m \overset{\checkmark}{=} 2^m. \tag{1}$$

Then U is by construction a compact set in $\mathbb{R}^4$ (finite product of closed bounded sets). Since L is an automorphism on $\mathbb{R}^4$ and in particular an homeomorphism on $\mathbb{R}^4$, it implies that $S = L^{-1}(U)$ is compact as well. $\checkmark$

claim. *S is convex and symmetric.*

Since $\det(\mathbb{Z}^4) = 1$, and (1) together with above claim, Minkowski's lattcie point theorem implies the existence of a $x = (x_1, \dots, x_m) \in \mathbb{Z}^4 \setminus \{0\}$, such that $x \in S$ and by deifinition of S the statement is proven. ✓

It is left to prove the claim.
We claim that U is convex. Indeed, let $x = (x_1, \dots, x_m), y = (y_1, \dots, y_m) \in U$. Then without loss of generality we may assume that $|x_i| \geq |y_i|$ for each $i \in \{1, \dots, m\}$. (otherwise replace x_i with y_i in the following) Then let $\lambda \in [0, 1]$. For each $i \in \{1, \dots, m\}$ it holds that

$$|(\lambda x + (1-\lambda)y)_i| \overset{\checkmark}{\leq} \lambda|x_i| + (1-\lambda)|y_i| \overset{\checkmark}{\leq} \lambda|x_i| + (1-\lambda)|x_i| = |x_i| \overset{\checkmark}{\leq} \delta_i,$$

so by construction of U, we get $\lambda x + (1-\lambda)y \in U$, thus U is convex. ✓ (I mean, U is a hypercube)
Now, let $x, y \in S$ and $\lambda \in [0, 1]$. It holds that $L(x), L(y) \in U$ and by linearity of L^{-1}, we get that

$$\lambda x + (1-\lambda)y \overset{\checkmark}{=} \lambda L^{-1}(L(x)) + (1-\lambda)L^{-1}(L(y)) \overset{\checkmark}{=} L^{-1}(\underbrace{\lambda L(x) + (1-\lambda)L(y)}_{\in U \text{ by convexity of } U}) \overset{\checkmark}{\in} L^{-1}(U) = S,$$

so convexity of S follows. ✓
Furthermore it holds that $L(-x) \in U$, since for any $x = (x_1, \dots, x_m) \in U$, it holds that $|-x_i| = |x_i| \leq \delta_i$, thus $-x \in L^{-1}(U) = S$ hence, S is symmetric and we can conclude. ✓ □

4

Aufgabe 3

Let $p \geq 2$ be a prime number. Then let $a, b \in \mathbb{Z}$ such that $a^2 + b^2 \equiv -1 \pmod{p}$. We denote $v_1 = (p, 0, 0, 0)^t, v_2 = (0, p, 0, 0)^t, v_3 = (a, b, 1, 0)^t, v_4 = (-b, a, 0, 1)^t \in \mathbb{R}^4$ and

$$\Lambda = v_1\mathbb{Z} + v_2\mathbb{Z} + v_3\mathbb{Z} + v_4\mathbb{Z}.$$

We denote $||\cdot||_2$ for the euclidian norm on $\mathbb{R}^4$.

(i)

t.s. $\Lambda \subseteq \mathbb{R}^4$ *is a lattice with* $\det(\Lambda) = p^2$.

Proof. Definition 4.6 for $\Lambda \subseteq \mathbb{R}^4$ being a lattice is equivalent to $\Lambda \subseteq \mathbb{R}^4$ being a free finitely generated $\mathbb{Z}$-modulewith $\operatorname{rank}(\Lambda) = 4$ which is generated by a basis of $\mathbb{R}^4$. (a subgroup of $\mathbb{R}^4$ must be abelian hence, has a $\mathbb{Z}$-modul structure on it and conversely a $\mathbb{Z}$-modul must be an abelian Group. Since generated from a basis of $\mathbb{R}^4$, it must be a subgroup of $\mathbb{R}^4$.) By construction Λ is a finitely generated $\mathbb{Z}$-modul. We consider the matrix

$$A = (v_1|v_2|v_3|v_4) = \begin{pmatrix} p & 0 & a & -b \\ 0 & p & b & a \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in Mat_4(\mathbb{R})$$

Then A is a trianglularmatrix and we get $\det(A) = p^2$. Since $p \neq 0$ it follows that the columns (v_1, v_2, v_3, v_4) are lineary independent hence, (v_1, v_2, v_3, v_4) forms a basis of $\mathbb{R}^4$ and it follows that $\Lambda \subseteq \mathbb{R}^4$ forms a lattice. In particular, this shows that $\det(\Lambda) = p^2$, since $\det(\Lambda) = \det(A)$ by definition and we can conclude. □

(ii)

t.s. *It holds that* $||v||_2^2 \in p\mathbb{Z}$ *for any* $v \in \Lambda$.

Proof. Let $v \in \Lambda$. Then there exist $z_1, z_2, z_3, z_4 \in \mathbb{Z}$ such that

$$v = z_1v_1 + z_2v_2 + z_3v_3 + z_4v_4.$$

Thus

$$v = \begin{pmatrix} z_1p + z_3a - z_4b \\ z_2p + z_3b + z_4a \\ z_3 \\ z_4 \end{pmatrix},$$

and we get

$$\begin{aligned}
||v||_2^2 &= (z_1p + z_3a - z_4b)^2 + (z_2p + z_3b + z_4a)^2 + z_3^2 + z_4^2 \\
&= z_1^2p^2 + z_3^2a^2 + z_4^2b^2 + 2z_1z_3pa - 2z_1z_4pb - 2z_3z_4ab \\
&\quad + z_2^2p^2 + z_3^2b^2 + z_4^2a^2 + 2z_2z_3pb + 2z_2z_4pa + 2z_3z_4ab + z_3^2 + z_4^2 \\
&= \underbrace{(z_1^2 + z_2^2)p^2}_{\equiv 0 \pmod{p}} + \underbrace{p(2a(z_1z_3 + z_2z_4) + 2b(z_2z_3 - z_1z_4))}_{\equiv 0 \pmod{p}} + z_3(a^2 + b^2) + z_4^2(a^2 + b^2) + \dots \\
&\equiv z_3^2 \underbrace{(a^2 + b^2)}_{\equiv -1 \pmod{p}} + z_4^2 \underbrace{(a^2 + b^2)}_{\equiv -1 \pmod{p}} + z_3^2 + z_4^2 \pmod{p} \\
&\equiv -z_3^2 - z_4^2 + z_3^2 + z_4^2 \pmod{p} \equiv 0 \pmod{p}.
\end{aligned}$$

It follows that $||v||_2^2 \in p\mathbb{Z}$ and we can conlcude. □

(iii)

t.s. *It holds that*

$$\Lambda \cap \{x \in \mathbb{R}^4 : ||x||_2^2 < 2p\} \supsetneq \{0\}.$$

Proof. We denote $S = \{x \in \mathbb{R}^4 : ||x||_2^2 < 2p\}$. Then it holds by (i) that $2p = \sqrt{4\det(\Lambda)}$, and it follows by construction that $S = B_r(0)$, a ball with radius r and centered at 0, where $r = \sqrt[4]{4\det(\Lambda)}$.

claim. $S = B_r(0)$ *is symmetric and convex.*

Since S is open in $\mathbb{R}^4$, it is Lebesgue measurable and it holds that the volume of any ball with radius r is $\frac{pi^2}{2}r^4$. We get

$$vol(S) = \frac{pi^2}{2}r^4 = \frac{pi^2}{2}4\det(\Lambda) = \underbrace{\pi^2}_{>3^2>2^3} 2\det(\Lambda) > 2^4\det(\Lambda).$$

Together with claim, Minikowskis Latticepoint Theorem provides us the existence of a lattice point $v \in S \cap (\Lambda \setminus \{0\})$. It follows that $\Lambda \cap \{x \in \mathbb{R}^4 : ||x||_2^2 < 2p\} \supseteq \{v, 0\} \supsetneq \{0\}$ and the statement ~~implies~~ follows.

It is left to prove the claim. Let $x, y \in B_r(0)$. Then $||x||_2, ||y||_2 < r$.
We define $m = \max\{||x||_2, ||y||_2\} < r$. Then it holds for any $\lambda \in [0, 1]$, that

$$||\lambda x + (1-\lambda)y||_2 \leq \lambda||x||_2 + (1-\lambda)||y||_2 \leq \lambda m + (1-\lambda)m = m < r$$

Thus S is convex. Furthermore it holds that $||-x||_2 = ||x||_2 < r$ so it is symmetric as well and we can conclude. □

(iv)

t.s. *There exist some integer* $u_1, u_2, u_3, u_4 \in \mathbb{Z}$, *such that* $p = u_1^2 + u_2^2 + u_3^2 + u_4^2$.

Proof. By definition it holds that $v_1, \ldots v_4 \in \mathbb{Z}^4$, ($v_i$ from (i)) thus Λ is a integer lattice, i.e. $\Lambda \subseteq \mathbb{Z}^4$ (as a set). Then by (iii) there exists a $u \in \Lambda$ such that $u \neq 0$ and $||u||_2^2 < 2p$. Let $u = (u_1, u_2, u_3, u_4) \in \Lambda$ be the representation of u in coordinates. Then $u_1, \ldots, u_4 \in \mathbb{Z}$ since Λ a integer lattice. And the arguments before imply that

$$0 < ||u||_2^2 = u_1^2 + u_2^2 + u_3^2 + u_4^2 < 2p \tag{2}$$

Then by (ii) it holds that $||u||_2^2 \in p\mathbb{Z}$. Hence, there exists a $z \in \mathbb{Z}$, such that $||u||_2^2 = u_1^2 + u_2^2 + u_3^2 + u_4^2 = pz$. Together with (2), we get that

$$0 < pz < 2p.$$

Thus $0 < z < 2$. But since z a integer, this implies that $z = 1$, which gives us

$$u_1^2 + u_2^2 + u_3^2 + u_4^2 = p$$

and we can conclude. □